

Task 4P

Solve the following set of problems using Python and submit the code file with extension .ipynb.

1. Download [Electrical Grid Stability Simulated Data](#) dataset and print the dimension of the dataset.
2. Classify the "Electrical Grid Stability Simulated Data" (target=stabf) available in the dataset using SVM with three different kernels. Select appropriate data splitting approach and performance metrics. Report the performances and the used model hyper-parameters.
3. Continue from question 2, use the same training and test set, use KNN and tune K on the training set, report the performance on the test set and the optimal K.
4. Continue from question 2, use the same training and test set, use Decision Tree and tune the depth of the tree on the training set, report the performance on the test set and the optimal depth.
5. Based on the model hyper-parameters used in question 2, 3 and question 4, share your understanding of hyper-parameters tuning in ML model development.

Assessment feedback

The results with comments will be released within 5 business days from the due date.

Referencing

You must correctly use the IEEE method in this assessment. See the [referencing guide](#).

Academic integrity, plagiarism and collusion

Plagiarism and collusion constitute extremely serious breaches of academic integrity. They are forms of cheating, and severe penalties are associated with them, including cancellation of marks for a specific assignment, for a specific unit or even exclusion from the course. If you are ever in doubt about how to properly use and cite a source of information refer to the referencing site above.

Plagiarism occurs when a student passes off as the student's own work, or copies without acknowledgement as to its authorship, the work of any other person or resubmits their own work from a previous assessment task.

Collusion occurs when a student obtains the agreement of another person for a fraudulent purpose, with the intent of obtaining an advantage in submitting an assignment or other work.

Work submitted may be reproduced and/or communicated by the university for the purpose of assuring academic integrity of submissions:
<https://www.deakin.edu.au/students/study-support/referencing/academic-integrity>.